**Student Course Management System Using MongoDB**

**Project Overview**

The Student Course Management System is a beginner-friendly database project developed using MongoDB. It manages student details, course enrollments, and learning progress.

**Technologies Used**

- MongoDB
- MongoDB Shell (mongosh)

**Features**

- Create a database and collections.
- Insert single and multiple student records.
- Retrieve all records and search by course or status.
- Update one or multiple student records.
- Delete individual records and clear a collection for practice.
- Use query operators such as $gt, $lt, $in, $and, $or, and $exists.

**Real-World Application**

Educational institutions and training centers can use a student management database to maintain student information, track course enrollments, and monitor course completion status.

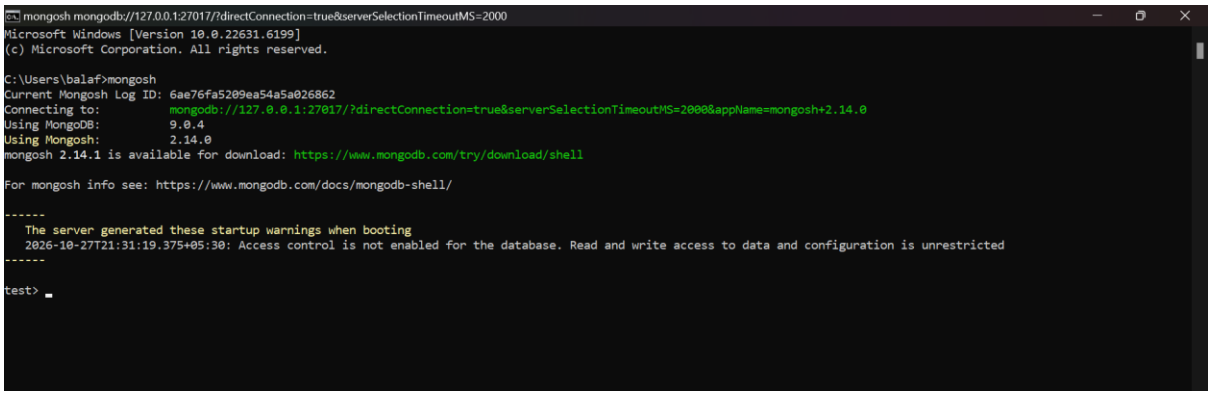
**Learning Outcomes**

Through this project, I learned MongoDB database and collection creation, CRUD operations, query operators, document structure, and basic database management.

**Conclusion**

This project helped me understand the fundamentals of MongoDB and how NoSQL databases can be used to manage structured information for real-world applications.

# Created database and installed



Created collection and inserting a data managemet of students

```
test> use StudentDB
switched to db StudentDB
StudentDB> db
StudentDB
StudentDB> db.students.insertOne({name:"Selva",age:"20",course:"MERN",status:"Completed"})
{
  acknowledged: true,
  insertedId: ObjectId('6ae7707f209ea54a5a026863')
}
StudentDB> db.students.find()
[
  {
    _id: ObjectId('6ae701b26e6019da2ca9d135'),
    age: '25',
    course: 'MERN',
    name: 'Balu',
    status: 'ongoing'
  },
  {
    _id: ObjectId('6ae702786e6019da2ca9d136'),
    name: 'vicky',
    age: '27',
    course: 'MERN',
    status: 'ongoing'
  },
  {
    _id: ObjectId('6ae702786e6019da2ca9d137'),
    name: 'siva',
    age: '25',
    course: 'java',
    status: 'completed'
  },
  {
    _id: ObjectId('6ae7707f209ea54a5a026863'),
    name: 'Selva',
    age: '20',
    course: 'MERN',
    status: 'Completed'
  }
]
StudentDB>
```

Showing all database

```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000
Microsoft Windows [Version 10.0.22631.6199]
(c) Microsoft Corporation. All rights reserved.

C:\Users\balaf>mongosh
Current Mongosh Log ID: 6ae76fa5209ea54a5a026862
Connecting to:          mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.14.0
Using MongoDB:          9.0.4
Using Mongosh:          2.14.0
mongosh 2.14.1 is available for download: https://www.mongodb.com/try/download/shell

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

------
   The server generated these startup warnings when booting
   2026-10-27T21:31:19.375+05:30: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
------

test> db
test
test> show databases
StudentDB  68.00 KiB
admin      40.00 KiB
bike       72.00 KiB
config     72.00 KiB
local      40.00 KiB
test> StudentDB
ReferenceError: StudentDB is not defined
test> use StudentDB
switched to db StudentDB
StudentDB>
```

Delete and find()

```
StudentDB> db.students.deleteOne({name:"siva"})
{ acknowledged: true, deletedCount: 1 }
StudentDB> db.students.find()
[
  {
    _id: ObjectId('6ae701b26e6019da2ca9d135'),
    age: '25',
    course: 'MERN',
    name: 'Balu',
    status: 'ongoing'
  },
  {
    _id: ObjectId('6ae702786e6019da2ca9d136'),
    name: 'vicky',
    age: '27',
    course: 'MERN',
    status: 'ongoing'
  },
  {
    _id: ObjectId('6ae7707f209ea54a5a026863'),
    name: 'Selva',
    age: '20',
    course: 'MERN',
    status: 'ongoing'
  }
]
StudentDB>
```

Filtered with query of atomic

```
StudentDB> db.students.find({course:"MERN"})
[
  {
    _id: ObjectId('6ae701b26e6019da2ca9d135'),
    age: '25',
    course: 'MERN',
    name: 'Balu',
    status: 'ongoing'
  },
  {
    _id: ObjectId('6ae702786e6019da2ca9d136'),
    name: 'vicky',
    age: '27',
    course: 'MERN',
    status: 'ongoing'
  },
  {
    _id: ObjectId('6ae7707f209ea54a5a026863'),
    name: 'Selva',
    age: '20',
    course: 'MERN',
    status: 'Completed'
  }
]
StudentDB>
```

Filetred by course name

```
StudentDB> db.students.find({course:"MERN"})
[
  {
    _id: ObjectId('6ae701b26e6019da2ca9d135'),
    age: '25',
    course: 'MERN',
    name: 'Balu',
    status: 'ongoing'
  },
  {
    _id: ObjectId('6ae702786e6019da2ca9d136'),
    name: 'vicky',
    age: '27',
    course: 'MERN',
    status: 'ongoing'
  },
  {
    _id: ObjectId('6ae7707f209ea54a5a026863'),
    name: 'Selva',
    age: '20',
    course: 'MERN',
    status: 'Completed'
  }
]
StudentDB> db.students.find({name:"Balu"})
[
  {
    _id: ObjectId('6ae701b26e6019da2ca9d135'),
    age: '25',
    course: 'MERN',
    name: 'Balu',
    status: 'ongoing'
  }
]
StudentDB>
```

Finding all

```
StudentDB> db.students.find()
[
  {
    _id: ObjectId('6ae701b26e6019da2ca9d135'),
    age: '25',
    course: 'MERN',
    name: 'Balu',
    status: 'ongoing'
  },
  {
    _id: ObjectId('6ae702786e6019da2ca9d136'),
    name: 'vicky',
    age: '27',
    course: 'MERN',
    status: 'ongoing'
  },
  {
    _id: ObjectId('6ae702786e6019da2ca9d137'),
    name: 'siva',
    age: '25',
    course: 'java',
    status: 'completed'
  },
  {
    _id: ObjectId('6ae7707f209ea54a5a026863'),
    name: 'Selva',
    age: '20',
    course: 'MERN',
    status: 'Completed'
  }
]
StudentDB>
```

Insertion operation

```
switched to db StudentDB
StudentDB> db
StudentDB
StudentDB> db.students.insertOne({name:"Selva",age:"20",course:"MERN",status:"Completed"})
{
  acknowledged: true,
  insertedId: ObjectId('6ae7707f209ea54a5a026863')
}
```

Below images are using query operators like $gt, $lt, $in, $and, $or, $exists

```
StudentDB> db.students.find({$and:[{age:{$gt:"20"}},{status:"ongoing"}]})
[
  {
    _id: ObjectId('6ae702786e6019da2ca9d136'),
    name: 'vicky',
    age: '27',
    course: 'MERN',
    status: 'ongoing'
  }
]
StudentDB> db.students.find({$or:[{status:"Completed"},{status:"ongoing"}]})
[
  {
    _id: ObjectId('6ae701b26e6019da2ca9d135'),
    age: '25',
    course: 'MERN',
    name: 'Balu',
    status: 'Completed'
  },
  {
    _id: ObjectId('6ae702786e6019da2ca9d136'),
    name: 'vicky',
    age: '27',
    course: 'MERN',
    status: 'ongoing'
  },
  {
    _id: ObjectId('6ae7707f209ea54a5a026863'),
    name: 'Selva',
    age: '20',
    course: 'MERN',
    status: 'ongoing'
  }
]
StudentDB>
```

```
StudentDB> db.students.find({$and:[{age:{$gt:"20"}},{status:"Completed"}]})
[
  {
    _id: ObjectId('6ae701b26e6019da2ca9d135'),
    age: '25',
    course: 'MERN',
    name: 'Balu',
    status: 'Completed'
  }
]
StudentDB> db.students.find({$and:[{age:{$gt:"20"}},{status:"ongoing"}]})
[
  {
    _id: ObjectId('6ae702786e6019da2ca9d136'),
    name: 'vicky',
    age: '27',
    course: 'MERN',
    status: 'ongoing'
  }
]
StudentDB>
```

```
StudentDB> db.students.find({age:{$exists:"true"}})
[
  {
    _id: ObjectId('6ae701b26e6019da2ca9d135'),
    age: '25',
    course: 'MERN',
    name: 'Balu',
    status: 'Completed'
  },
  {
    _id: ObjectId('6ae702786e6019da2ca9d136'),
    name: 'vicky',
    age: '27',
    course: 'MERN',
    status: 'ongoing'
  },
  {
    _id: ObjectId('6ae7707f209ea54a5a026863'),
    name: 'Selva',
    age: '20',
    course: 'MERN',
    status: 'ongoing'
  }
]
StudentDB>
```

```
StudentDB> db.students.find({age:{$gt:"25"}})
[
  {
    _id: ObjectId('6ae702786e6019da2ca9d136'),
    name: 'vicky',
    age: '27',
    course: 'MERN',
    status: 'ongoing'
  }
]
StudentDB>
```

```
]
StudentDB> db.students.find({age:{$gt:"25"}})
[
  {
    _id: ObjectId('6ae702786e6019da2ca9d136'),
    name: 'vicky',
    age: '27',
    course: 'MERN',
    status: 'ongoing'
  }
]
StudentDB> db.students.find({age:{$lt:"25"}})
[
  {
    _id: ObjectId('6ae7707f209ea54a5a026863'),
    name: 'Selva',
    age: '20',
    course: 'MERN',
    status: 'ongoing'
  }
]
StudentDB>
```

```
StudentDB> db.students.find({course:{$in:["MERN"]}})
[
  {
    _id: ObjectId('6ae701b26e6019da2ca9d135'),
    age: '25',
    course: 'MERN',
    name: 'Balu',
    status: 'ongoing'
  },
  {
    _id: ObjectId('6ae702786e6019da2ca9d136'),
    name: 'vicky',
    age: '27',
    course: 'MERN',
    status: 'ongoing'
  },
  {
    _id: ObjectId('6ae7707f209ea54a5a026863'),
    name: 'Selva',
    age: '20',
    course: 'MERN',
    status: 'ongoing'
  }
]
StudentDB>
```

UpdateOne and UpatedMany

